

LECTURE NOTES

L^AT_EX Lecture Note Template

Lecture 01 (Week 1)

Template Overview

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Contents

1	Introduction	1
2	Callout Boxes	1
3	Figures and Tables	2
4	Lists	3
5	Mathematics	3
6	Text Formatting	3
6.1	Font Styles	3
6.1.1	English	3
6.1.2	ภาษาไทย	3
6.1.3	Sample Text	3
6.2	Font Size	4
7	Code Snippets	4

1 Introduction

This document is Worralop Srichainont's personal $\text{\LaTeX}$ template for lecture summaries and study notes. The source is published in the [GitHub repository](#). It supports a cover page with lecture and week details, a running header/footer, a table of contents, a small set of callout boxes, clean figures and tables, and syntax-highlighted code snippets via `minted`. It also fully supports ภาษาไทย (Thai language) using the THSarabunPSK font. You are free to use and modify this template without any restrictions.

Note: file layout

The template is split into three files: `preamble.tex` (packages and styling you rarely touch), `metadata.tex` (the fields you edit for every new lecture: lecture number, week, topic, last updated), and `template.tex` (the actual content, this file). For graded assignments, use the sibling `assignment/` folder instead, which has its own independent cover and styling.

Before compiling this document, install the Pygments library required by the `minted` package:

```
1 pip install Pygments
2
```

For your $\text{\LaTeX}$ editor, [Overleaf](#) is the easiest online option. If you prefer to work locally, install $\text{\LaTeX}$ via [MikTeX](#) and use an editor such as [Visual Studio Code](#) (with the LaTeX Workshop extension) or [TeXMaker](#).

To compile, select **XeLaTeX** as the compiler, or run:

```
1 xelatex -shell-escape -interaction=nonstopmode template.tex
2
```

2 Callout Boxes

A small, muted set of callout boxes is available for flagging important content without cluttering the page.

Definition

A **callout box** is a visually distinct block used to draw attention to a specific piece of content, such as a definition, a worked example, or a warning, separate from the

surrounding body text.

Example

`\begin{examplebox}` renders this box. Pass an optional argument to change the title, e.g. `\begin{examplebox}[Example: recursion]`.

Key Takeaway

Use callout boxes sparingly. Reserve them for content you would circle in a printed handout, not for every paragraph, so they keep their emphasis.

3 Figures and Tables

Figures use `graphicx` with `caption` for numbering and `cleveref` for cross references, and tables use `booktabs` rules instead of a full grid, so both stay clean on the page.

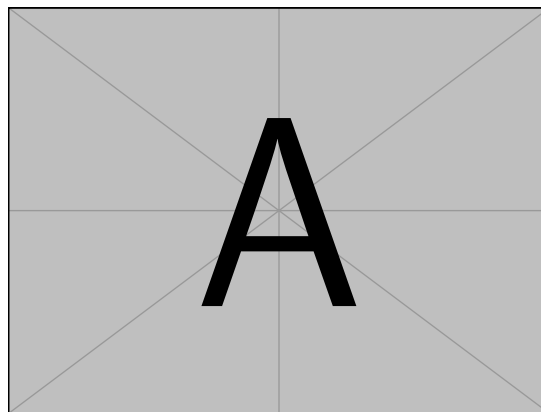


Figure 1: Placeholder figure. Replace `example-image-a` with your own image path.

Figure 1 shows a placeholder figure; Table 1 shows a placeholder table.

Table 1: Sample weekly topic outline using `booktabs` rules.

Week	Lecture	Topic
1	01	Template Overview
2	02	Sectioning & Lists
3	03	Math & Figures

4 Lists

`enumitem` controls list spacing globally, keeping lists airy but still compact enough to scan.

1. First key point from the lecture.
2. Second key point, with a supporting breakdown:
 - Supporting detail one.
 - Supporting detail two.
3. Final takeaway.

5 Mathematics

Numbered displays can be cross-referenced with `cleveref`. For example, Equation (1) gives the quadratic formula.

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \tag{1}$$

6 Text Formatting

6.1 Font Styles

L^AT_EX has various font styles, just like other document editors such as Microsoft Word.

6.1.1 English

6.1.2 ภาษาไทย

6.1.3 Sample Text

In Bangkok, **everywhere you look** is amazing food. To sound like a local, say *bhed nid noi* (เผ็ดนิดหน่อย), meaning *a little bit spicy*. Navigate the city using **Google Maps** or the BTS Skytrain (รถไฟฟ้า). Wherever you go, remember: Sawatdee krap (สวัสดีครับ)!

Command	Example
<code>text</code>	Normal text
<code>\textbf{text}</code>	Bold text
<code>\textit{text}</code>	<i>Italic text</i>
<code>\emph{text}</code>	<i>Emphasized</i>
<code>\underline{text}</code>	<u>Underline</u>
<code>\textsc{text}</code>	SMALL CAPS
<code>\texttt{text}</code>	Typewriter family

Command	Example
<code>text</code>	ข้อความ
<code>\textbf{text}</code>	ข้อความ
<code>\textit{text}</code>	<i>ข้อความ</i>
<code>\emph{text}</code>	<i>ข้อความ</i>
<code>\underline{text}</code>	<u>ข้อความ</u>

6.2 Font Size

Font sizes are relative to the size declared in `\documentclass`.

Command	Example
<code>{\small text}</code>	Small
<code>text</code>	Normal size
<code>{\large text}</code>	large
<code>{\Large text}</code>	Large
<code>{\LARGE text}</code>	LARGE
<code>{\huge text}</code>	huge

7 Code Snippets

Code blocks use minted through a `tcolorbox` wrapper, styled as a GitHub-dark listing.

example.py

```
1 def greet(name: str) -> str:
2     return f"Sawatdee, {name}!"
3
```

```
4 print(greet("world"))  
5
```